



INNOVATE. DEVELOP. DELIVER.

Building Smart Solutions,
Delivering Real Impact.

PROJECT TITLE

CivicSense AI

Issue Reporting and Validation System

An AI-powered platform for intelligent civic issue detection, validation, tracking, and real-time resolution.



Founder

Chandrasekhar Uppu

B.Tech



Passionate About

Building impactful software solutions that solve real-world problems.



B.Tech Expertise

Strong foundation in Computer Science and problem-solving.



Technical Skills

Proficient in Full-Stack Development, AI/ML, and Data Analytics.



Project Development

End-to-end development of scalable, efficient, and user-centric solutions.



Vision

To leverage technology for better governance and smarter communities.

0110100 0101
0100101 0101
0100100 1011
1101101 01
1101101
0101011
0100011
010001
1111
011
01
0



INNOVATE. DEVELOP. DELIVER.

Building Smart Solutions,
Delivering Real Impact.

TEAM DETAILS

Meet Our Team

Together we build intelligent solutions for a better tomorrow.



**UPPU
CHANDRASEKHAR**

FOUNDER

Leads the vision and strategy. Drives innovation and project direction.



**GOGU
NAGESWARI**

FRONTEND DEVELOPER

Designs and develops responsive and intuitive user interfaces.



**GURRAM
JYOTHI**

BACKEND DEVELOPER

Builds and maintains robust server-side logic and database systems.



**ADDALA
TEJASWINI**

**API HANDLER &
WEB DEVELOPER**

Integrates APIs and develops dynamic web functionalities.

I. ABSTRACT

CivicSense AI is an intelligent web + mobile platform that empowers citizens and city authorities to collaboratively detect, report, and resolve civic issues such as potholes, garbage overflow, and streetlight failures. By combining AI-powered image detection, predictive analytics, and real-time dashboards, it transforms traditional complaint systems into a proactive, data-driven urban management solution.

II. PROBLEM STATEMENT

Urban areas face persistent civic issues due to delayed complaint reporting, manual categorization and prioritization, lack of predictive maintenance, and inefficient citizen–authority communication. These problems result in slow responses, resource wastage, and citizen dissatisfaction.

III. PROPOSED SOLUTION (MVP)

1. Citizen App

Allows users to upload photos or voice reports. AI auto-identifies issue type and location.

2. AI Detection

Computer vision models classify issues from images; NLP models process text and voice inputs.

3. Predictive Analytics

ML predicts future civic issues using complaint, weather, and demographic data.

4. Authority Dashboard

Provides heatmaps, priorities, and analytics for data-driven decision-making.

5. Technology Stack

React.js · Node.js · MongoDB · Flask · TensorFlow CNN · JWT Authentication

Keywords: Civic Issue Detection · Deep Learning · CNN · Predictive Analytics · Urban Management · AI Image Validation · MERN Stack · Flask API

AI-Powered CivicSense System — Complete Workflow

System Design · Data Flow · Architecture Layers

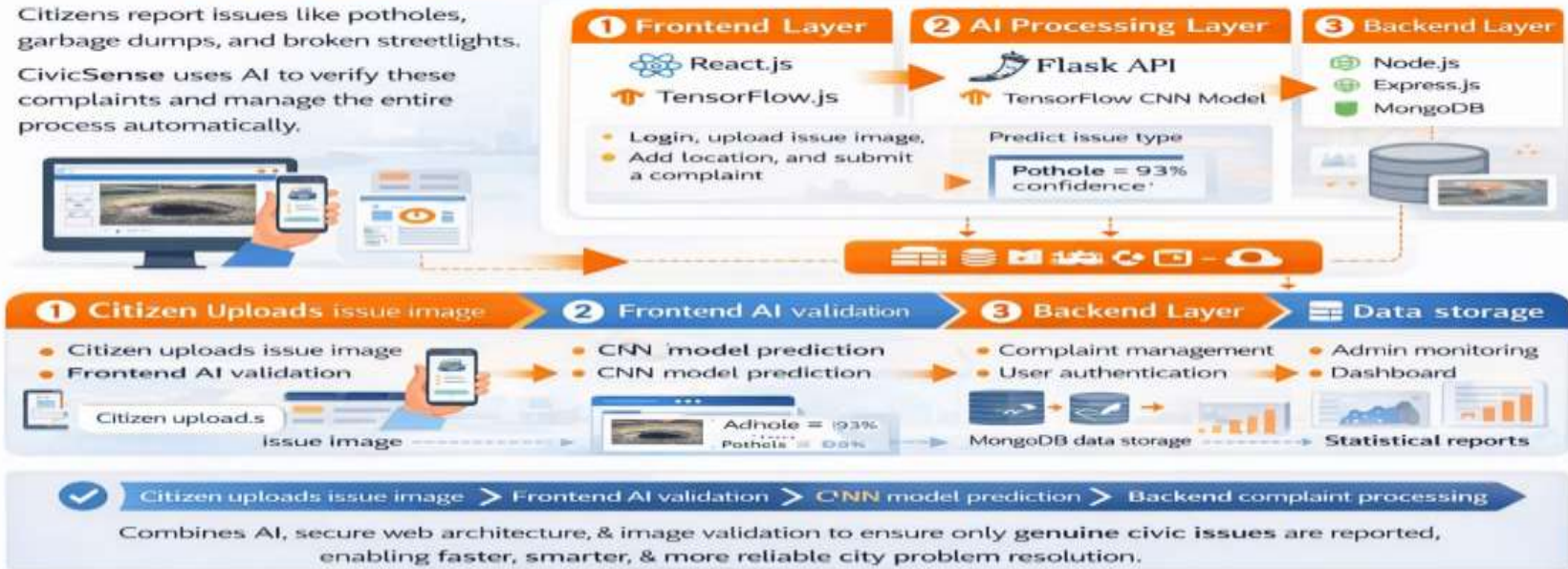
IV. SYSTEM WORKFLOW

AI-Powered CivicSense System – Complete Workflow

AI-powered platform for fast and reliable civic issue reporting.

Citizens report issues like potholes, garbage dumps, and broken streetlights.

CivicSense uses AI to verify these complaints and manage the entire process automatically.

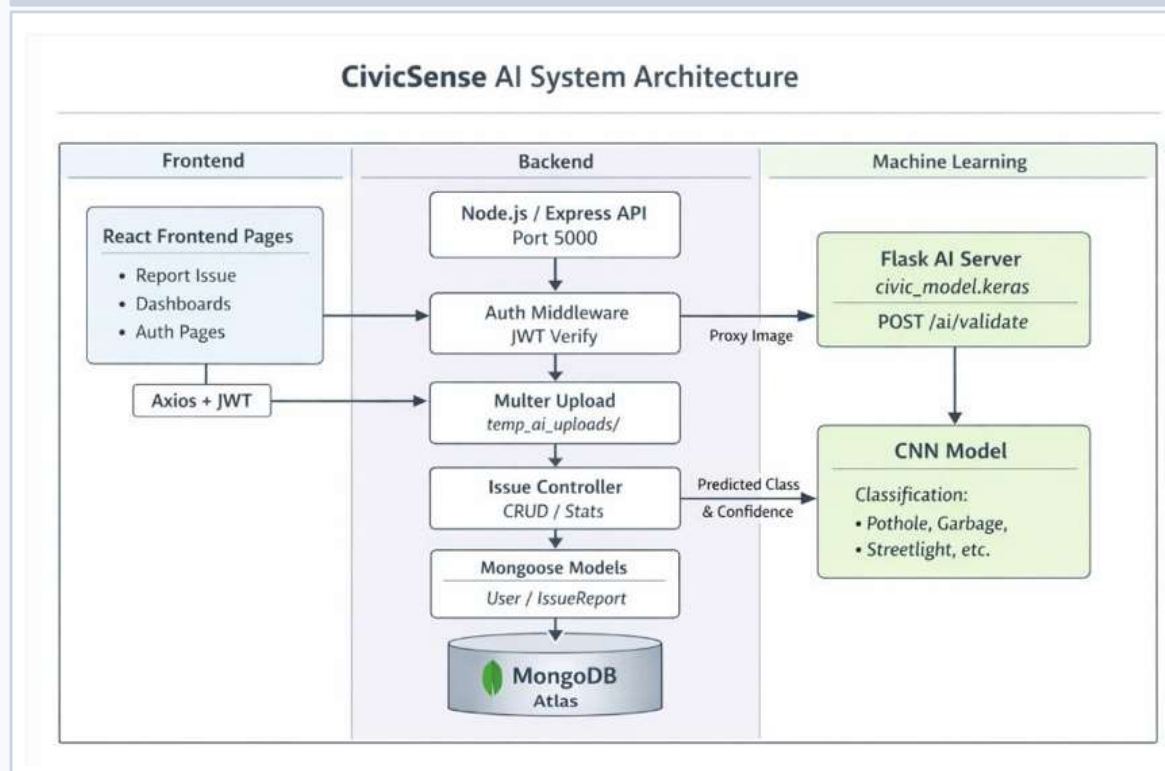




AI-Powered CivicSense System :CivicSense AI vs Existing Systems

Feature	Traditional	Digital Portals	CivicSense AI ✓
Reporting Method	Phone / Walk-in	Manual text forms	AI-validated image upload
Image Validation	✗ None	✗ None	✓ YOLOv8 Deep Learning
Location Capture	Manual / Vague	Manual entry	✓ GPS + Reverse Geocoding
Authentication	✗ None / Basic	Username + Password	✓ JWT + RBAC + Bcryptjs
Admin Dashboard	✗ None	Basic list view	✓ Analytics, filters, comments
Citizen Tracking	✗ No feedback	Limited status	✓ Real-time status badge
Scalability	Very Low	Low	✓ Cloud-deployed, horizontally scalable

VI. SYSTEM ARCHITECTURE



Frontend Layer

- React.js SPA
- TensorFlow.js client-side validation
- Axios + JWT authentication
- Citizen app & admin dashboard

Backend Layer

- Node.js / Express API (Port 5000)
- JWT Middleware authentication
- Multer file upload handler
- Mongoose ORM · MongoDB Atlas

ML / AI Layer

- Flask AI Server (Python)
- CNN Model (civic_model.keras)
- POST /ai/validate endpoint
- Classes: Pothole, Garbage, Streetlight

Testing & Validation Results

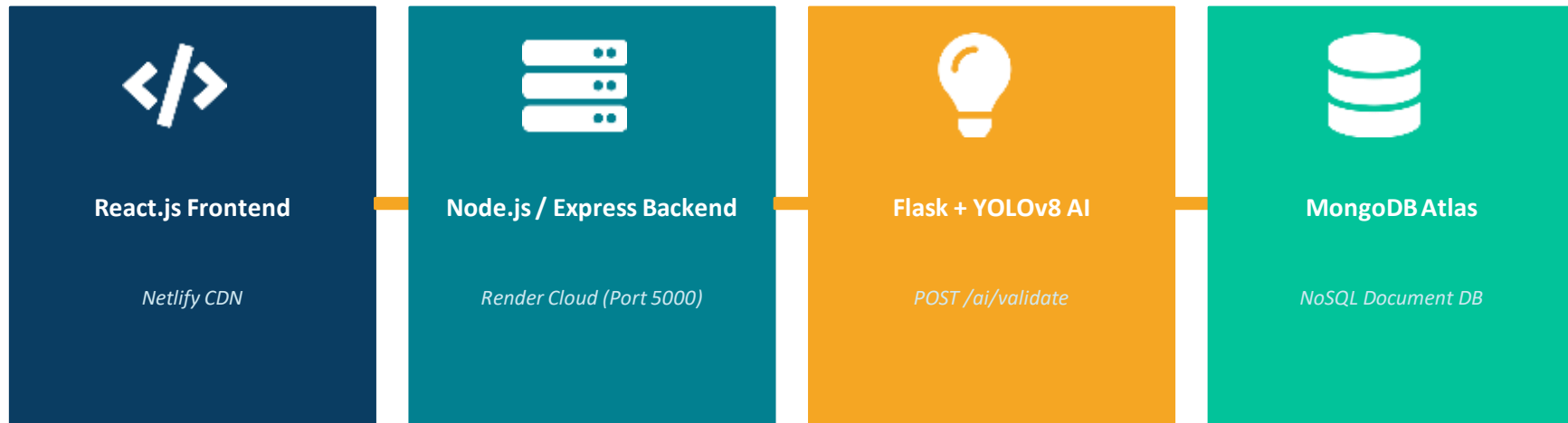
100%

Pass Rate

14 / 14
Test Cases

TC#	Scenario	Result
TC01-02	Pothole / Garbage image AI match	✔ Passed
TC03	Wrong image vs selected type → mismatch warning	✔ Passed
TC05	Invalid file type (PDF) → rejection	✔ Passed
TC07-08	Valid signup (201) / Duplicate email (400)	✔ Passed
TC09-10	Valid login (200) / Wrong password (401)	✔ Passed
TC11-12	Admin login / Citizen token on admin route (403)	✔ Passed

Four-Tier System Architecture



Citizen submits photo → Backend proxies to Flask AI → YOLOv8 returns class + confidence → Match validated → Report stored in MongoDB

JWT + RBAC two-tier authentication | HTTPS enforced | OWASP Top 10 hardened | Bcryptjs password hashing



CivicSense AI

Intelligent Civic Issue Management System

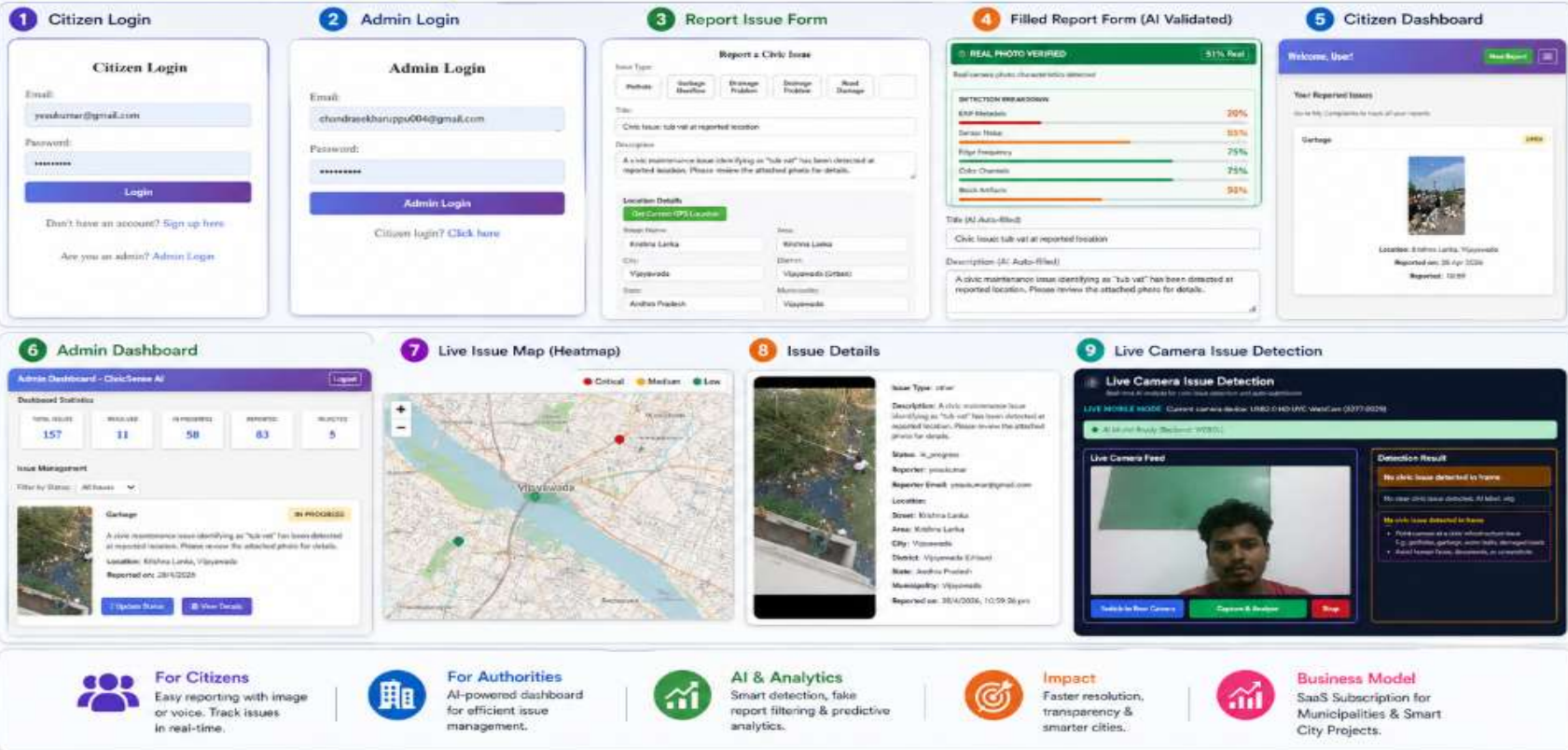
AI-Powered • Smart Detection • Real-Time Tracking • Better Cities



Our Mission

To empower citizens and authorities with AI-driven technology for a cleaner, smarter and more responsive city.

APPLICATION INTERFACE & USER FLOW



1000

✗ SCENARIO 1 — Image Mismatch REJECTED

Upload Image:

Screenshot 2025-11-19 102210.png



Issue Not Found!

Top predictions: web site, website, internet site, site (42.8%), newspaper (22.7%), meeting clock (2.8%)

X Image does not match issue type

Top predictions: web site, website, internet site, site (42.8%), newspaper (22.7%), meeting clock (2.8%)

Location obtained successfully

Submit Report

User Action: Selects Garbage Overflow type but uploads a screenshot (website image).**CNN Prediction:** Website (42.9%), Envelope (22.7%), Analog Clock (1.8%)**System Result:** Image does not match issue type — upload BLOCKED.

✓ SCENARIO 2 — Image Validated ACCEPTED

Upload Image:

garbage.png

☒ Image matches ixare type

Detecter: ashcan, trash can, garbage can, wastebbin, ash bin, ash-bin, ashbin, dustbin, trash barrel, trash bin (49.9%), ashcan, trash can, garbage can, wastebbin, ash bin, ash-bin, ashbin, dustbin, trash barrel, trash bin (49.9%), ashcan, trash can, garbage can, wastebbin, ash bin, ash-bin, ashbin, dustbin, trash barrel, trash bin (49.9%), ashcan, trash can, garbage can, wastebbin, ash bin, ash-bin, ashbin, dustbin, trash barrel, trash bin (49.9%), ashcan, trash can, garbage can, wastebbin, ash bin, ash-bin, ashbin, dustbin, trash barrel, trash bin (49.9%)

Location obtained successfully

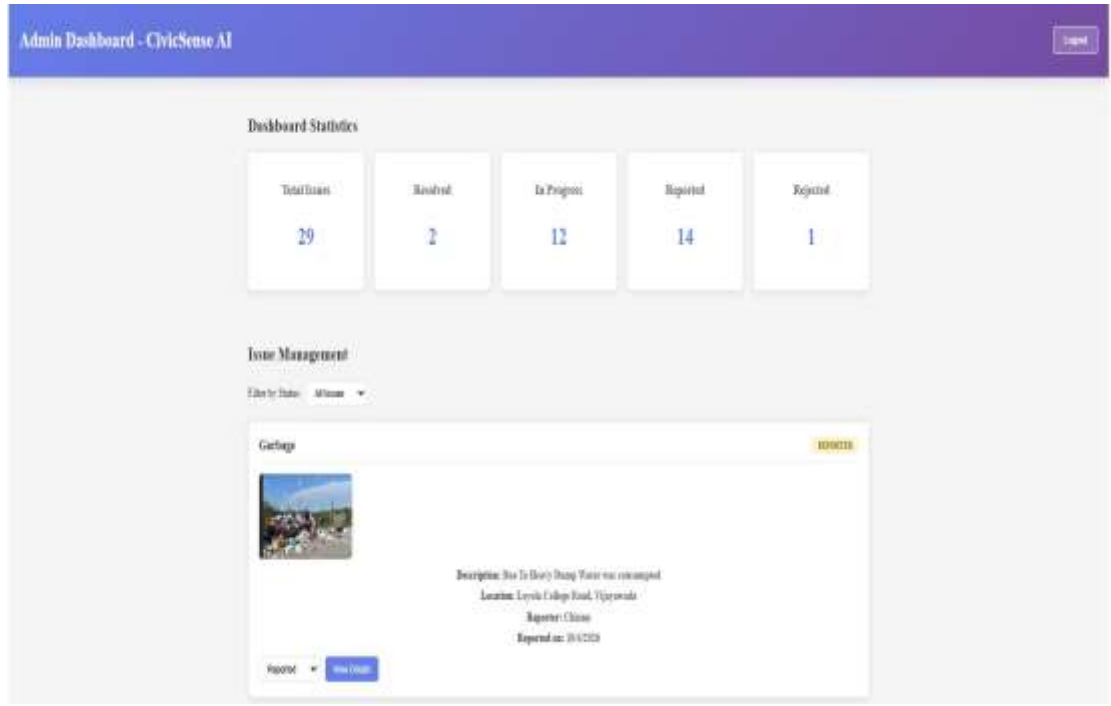
User Action: Selects Garbage Overflow and uploads actual garbage.png image. **CNN Prediction:** Ash-bin, Trash can, Garbage can, Dustbin (49.9%) — match confirmed! **System Result:** Image matches issue type — report SUBMITTED successfully.

Issue Tracking & Admin Management Dashboard

Real-Time Monitoring · Status Management · Issue Details

IX. ISSUE TRACKING & REAL-TIME ADMIN MONITORING

Admin Dashboard — Live Statistics & Issue Management



29

Total Issues

2

Resolved

12

In Progress

14

Reported

Issue Detail View — Admin Panel

Issue Details

Issue ID: 123456789



Description: Due To Heavy Dump Waste was accumulated

Reporter: Chitra

Reported on: 20/02/2025

Location: Loyola College Road, Vijayawada

Reporter: Chitra

Reported on: 20/02/2025

Status: In Progress

Assigned to: Chitra

Updated on: 20/02/2025

Reported on: 20/02/2025

Reported on: 20/02/2025

Reported on: 20/02/2025

Conclusion

CivicSense AI delivers a production-ready platform that demonstrates how AI-driven image validation, geospatial reporting, and transparent citizen-administrator communication can meaningfully transform urban infrastructure management.

- ✓ YOLOv8 AI validation reduces irrelevant complaints before they reach administrators
- ✓ 100% API test pass rate across 14 functional scenarios
- ✓ GPS + Nominatim geocoding delivers precise, human-readable issue locations
- ✓ JWT + RBAC security architecture enforces role boundaries (403/401)
- ✓ Four-tier MERN + Flask architecture confirms cloud scalability and deployment readiness

THANK YOU